EXP No: 8a NMAP to Discover Live Hosts Using
Date: 10/9/26 Nmap Scans (ARP, ICMP, TCP/UDP)
on the TryHackMe Platform

AIM:-

To discover live hosts using Nmap through ARP, ICMP, TCP and UDP scanning techniques on the TryHackMe platform.

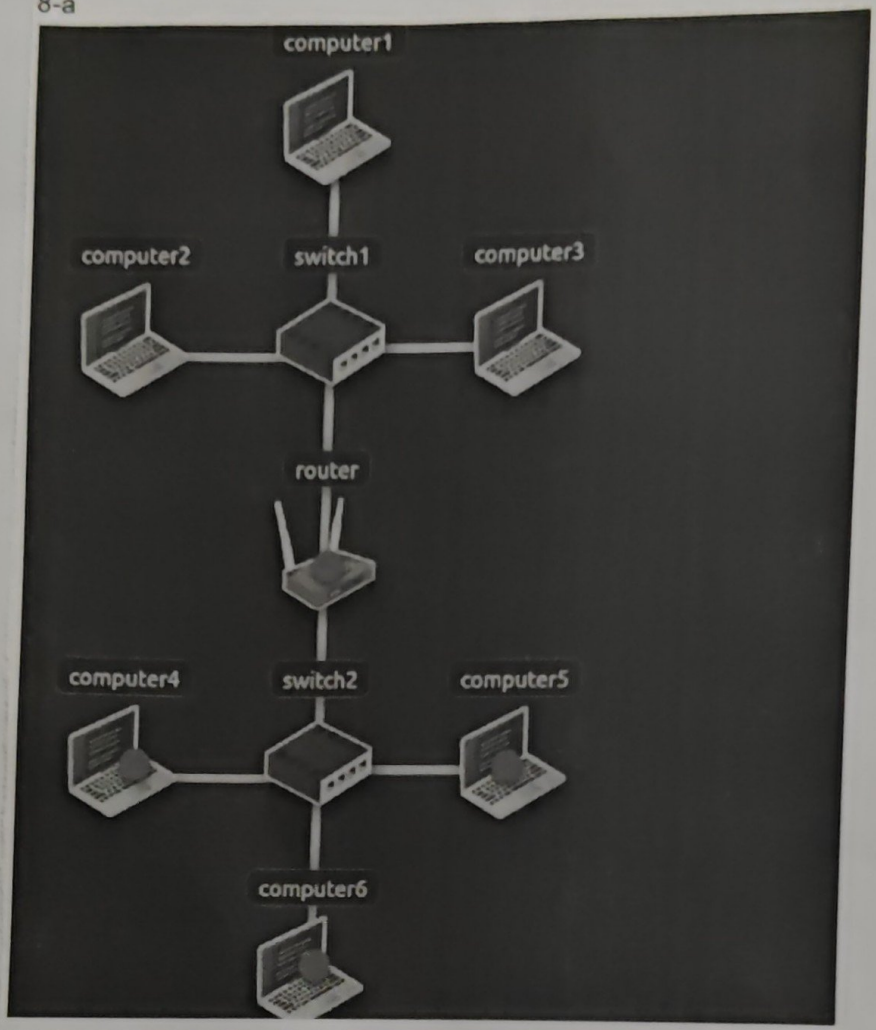
Procedure:

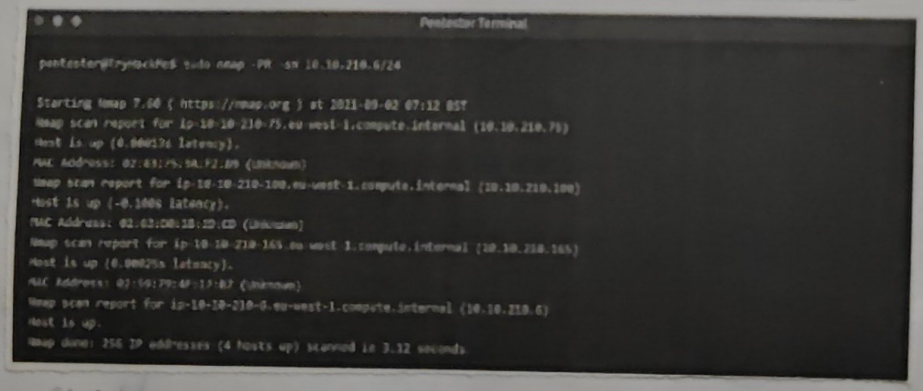
1) Open the TryHackMe Nmap Live Host Discovery room.
2) Study the network structure and identify the target IP addresses or subnet.
3) Specify the targets using an IP address, range, list or subnet
4) Use ARP scanning to identify live hosts on the same subnets.
5) Use ICMP Echo, Timestamp and Address mask requests for host discovery.
6) Use TCP SYN Ping and TCP ACK ping to check whether the target hosts are active
7) Use UDP ping to identify hosts through UDP and ICMP responses.
8) Use -Sn when only live host discovery is required without port scanning
9) Use reverse DNS lookup when host names of discovered systems are required
10) Record the responses and identify the live hosts

Important Commands:-

→ nmap -PR -Sn TARGETS - ARP host discovery
→ nmap -PE -Sn TARGETS - ICMP Echo discovery
→ nmap -PP -Sn TARGETS - ICMP Timestamp discovery
→ nmap -PM -Sn TARGETS - ICMP Address mask discovery
→ nmap -PS 23 TARGETS - TCP SYN Ping on Telnet Port 23

8-a


computer1
computer2
switch1
computer3
router
computer4
switch2
computer5
computer6


Pentester Terminal
pentester@TryHackMe$ sudo nmap -PR -sn 10.10.210.6/24
Starting Nmap 7.60 (https://nmap.org) at 2021-09-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.000174 latency).
MAC Address: 02:69:75:9A:F2:09 (Unknown)
Nmap scan report for ip-10-10-210-100.eu-west-1.compute.internal (10.10.210.100)
Host is up (-0.100s latency).
MAC Address: 02:62:DB:1B:2D:CD (Unknown)
Nmap scan report for ip-10-10-210-165.eu-west-1.compute.internal (10.10.210.165)
Host is up (0.00025s latency).
MAC Address: 02:59:79:4F:17:B7 (Unknown)
Nmap scan report for ip-10-10-210-6.eu-west-1.compute.internal (10.10.210.6)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.12 seconds

→ nmap -PA TARGETS — TCP ACK Ping
→ nmap -R TARGETS - Reverse DNS lookup
→ nmap -n TARGETS - Disable DNS resolution

ARP is useful for host's on the same subnet, while ICMP, TCP and UDP methods can be used for host discovery in different network situations.

Output :-

The following observations were obtained during the experiment :-

1. ARP Request was used to identify active hosts on the same subnet.
2. ARP Response indicated that the target host was active
3. ICMP Echo Request and Echo Reply were used for Ping-based discovery
4. ICMP Timestamp discovery used -PP.
5. ICMP Address Mask discovery used - PM.
6. ICMP Echo discovery used -PE
7. TCP SYN Ping used - PS
8. TCP ACK Ping used - PA
9. UDP Ping could identify a host through and ICMP Port unreachable response.
10. Reverse DNS lookup used -R
11. The -Sn option was used for host discovery without Port Scanning.

Result :-

Thus, the experiment was successfully completed. The different nmap give host discovery techniques using ARP, ICMP, TCP and UDP were studied and used to identify activehosts.

Ex No. 8b
Date: 10/9/26

Configuration of wireless LAN using CISCO Packet Tracer

AIM:

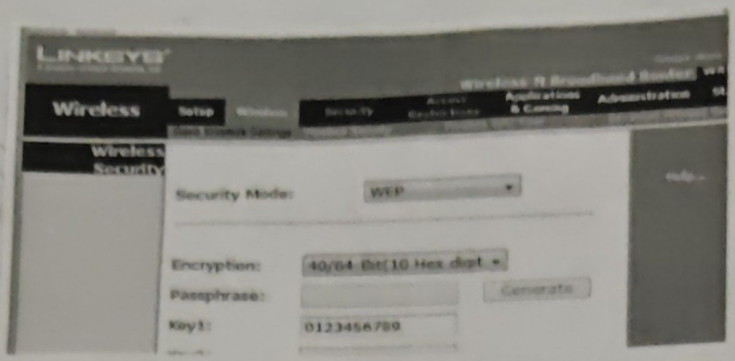
To design and configure a wireless LAN in using CISCO Packet Tracer.

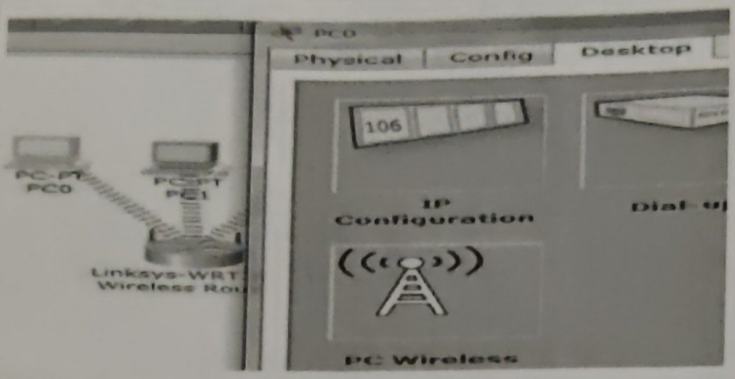
PROCEDURE:-

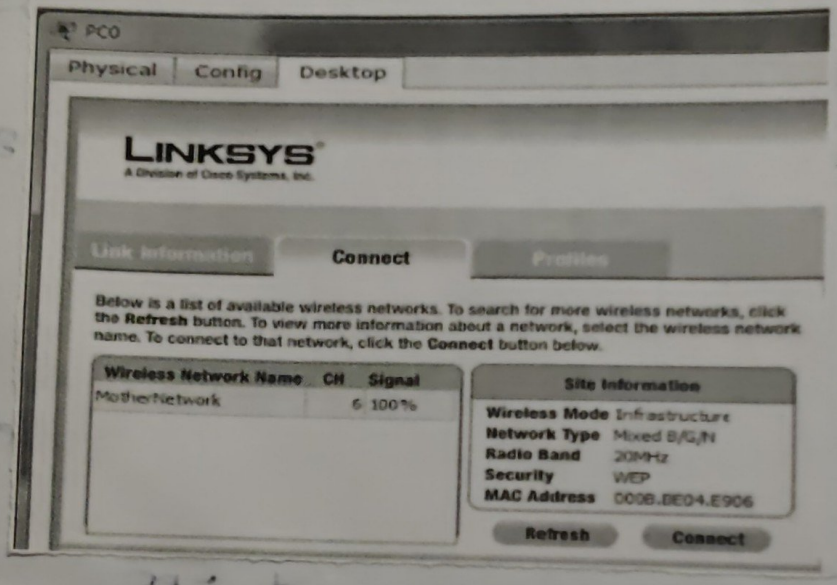
1. Create the topology
   - ↳ open Cisco Packet Tracer
   - ↳ Place one Linksys wireless router and 3 PC's : PC0, PC1 and PC2
   - ↳ Connect the PC's to the wireless routes.
2. Configure the wireless Router
   - ↳ click on the wireless Routes
   - ↳ Select the Administration tab
   - ↳ Set the username and Password as :
     - ★ Username : admin
     - ★ Password : admin
   - ↳ click Save settings.
3. Configure the SSID
   - ↳ Go to the wireless tab
   - ↳ change the default SSID to : Mother Network
   - ↳ Save the settings
4. Configure static IP Addresses

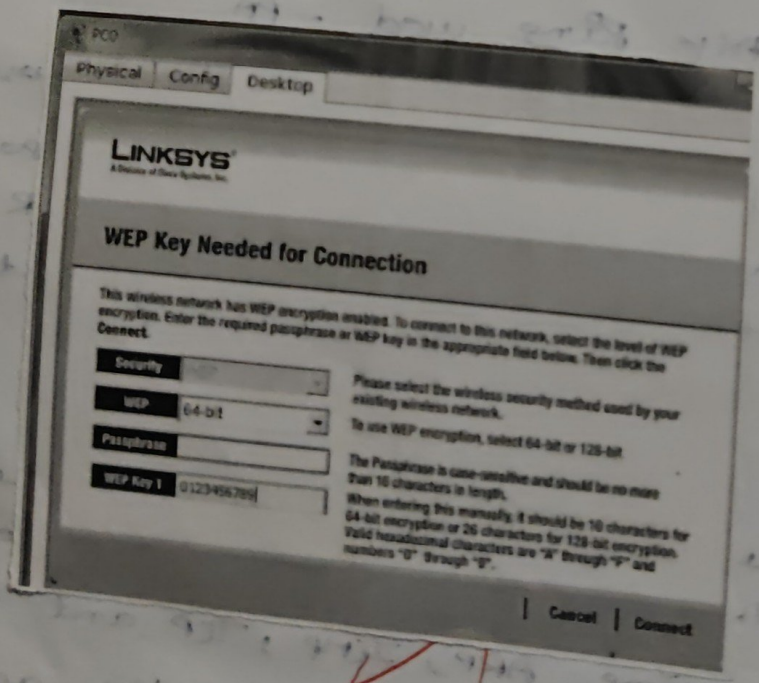
PC → Desktop → IP configuration → static

Device	IP Address	Subnet Network	Default Gateway
Router	192.168.0.1	255.255.255.0	—
PC0	192.168.0.2	255.255.255.0	192.168.0.1
PC1	192.168.0.3	255.255.255.0	192.168.0.1
PC2	192.168.0.4	255.255.255.0	192.168.0.1

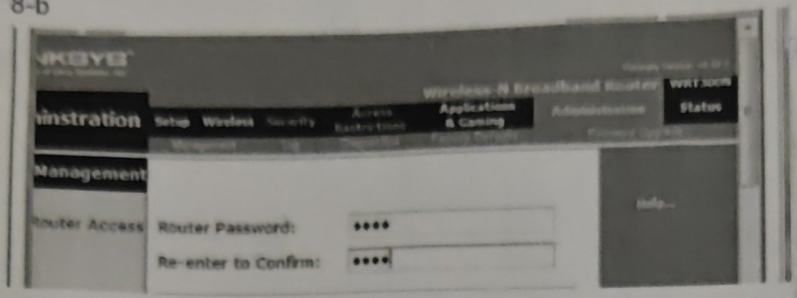

LINKSYS
Wireless
Setup
Security
Application & Gaming
Administration
Wireless Security
Security Mode:
WEP
Encryption:
40/64-Bit(10 Hex digit
Passphrase:
Generate
Key1:
0123456789

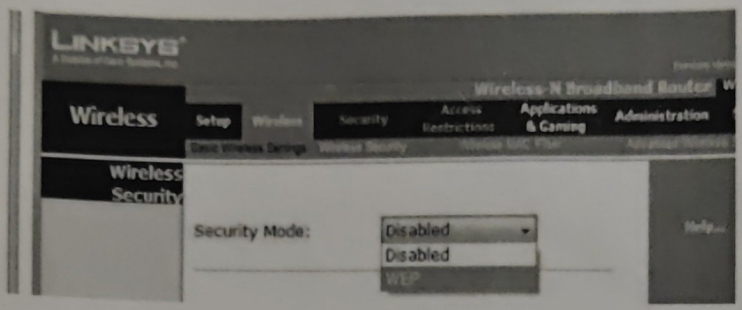

PC0
Physical
Config
Desktop
PC-PT PC0
PC-PT PC1
Linksys-WRT Wireless Ro
106
IP Configuration
Dial-up
PC Wireless

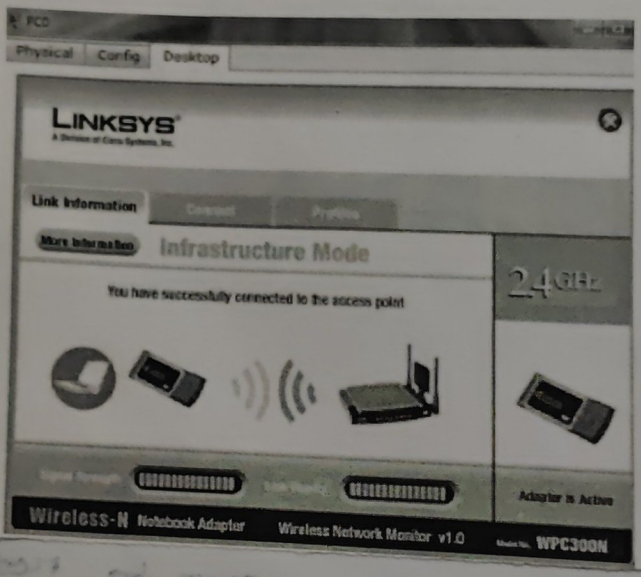

PC0
Physical
Config
Desktop
LINKSYS
A Division of Cisco Systems, Inc.
Link Information
Connect
Profiles
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
CH
Signal
MotherNetwork
6
100%
Site Information
Wireless Mode Infrastructure
Network Type Mixed B/G/N
Radio Band 20MHz
Security WEP
MAC Address 0008.DE04.E906
Refresh
Connect


PC0
Physical
Config
Desktop
LINKSYS
WEP Key Needed for Connection
This wireless network has WEP encryption enabled. To connect to this network, select the level of WEP encryption. Enter the required passphrase or WEP key in the appropriate field below. Then click the Connect.
Security
WEP
64-Bit
Passphrase
WEP Key 1
0123456789
Please select the wireless security method used by your existing wireless network.
To use WEP encryption, select 64-bit or 128-bit.
The Passphrase is case-sensitive and should be no more than 16 characters in length.
When entering this manually, it should be 10 characters for 64-bit encryption or 26 characters for 128-bit encryption. Valid hexadecimal characters are "A" through "F" and numbers "0" through "9".
Cancel
Connect

8-b


Wireless-N Broadband Router
Status
Management
Router Access
Router Password:
Re-enter to Confirm:


LINKSYS
Wireless-N Broadband Router
Wireless
Setup
Security
Access Restrictions
Applications & Gaming
Administration
Wireless Security
Security Mode:
Disabled
Disabled
WEP


PC0
Physical
Config
Desktop
LINKSYS
Link Information
More Information
Infrastructure Mode
2.4GHz
You have successfully connected to the access point
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
WPC300N

5. Configure wireless security
   - ↳ select wireless security
   - ↳ Set the security Mode to WEP.
   - ↳ Set key 1 as : 0123456789
   - ↳ click save settings.

6. Connect PC0 to the wireless Network
   - ↳ click PC's
   - ↳ Select Desktop → PC wireless
   - ↳ Select the connect tab
   - ↳ click refresh
   - ↳ Select Mother Network
   - ↳ click Connect
   - ↳ when prompted for the security key. enter : 0123456789
   - ↳ click connect

7. verify the connection
8. Connect PC1 and PC2
9. verify the wireless LAN after connecting all 3

Result :-

The configuration of wireless LAN has been successfully completed.